

THE HIGH-EXPONENT DUNFORD–PETTIS QUESTION FOR $A_2^r(\mathbb{Z})$

AGENT LANE 04

STATUS

This packet gives a scoped full answer to the $G = \mathbb{Z}$, $p = 2$ part of a question in Granirer [1]. The source asks whether $A_p^r(G)$ has the Dunford–Pettis property when $r > \max\{p, p'\}$, even for $G = \mathbb{Z}$ and $p = 2$. The answer is:

$A_2^r(\mathbb{Z})$ fails the Dunford–Pettis property for every $1 < r < \infty$,

and in particular for every finite $r > 2$. The endpoint $r = \infty$ is different: by definition $A_2^\infty(\mathbb{Z}) = A_2(\mathbb{Z})$, which is isometric to $L^1(\mathbb{T})$ and therefore has the Dunford–Pettis property.

SOURCE QUESTION

After proving the DPP failure for discrete G in the range $1 < r \leq \max\{p, p'\}$, the source asks:

It is not clear to us if $A_p^r(G)$ has the DPP if $r > \max\{p, p'\}$ in case G is discrete, or even if $G = \mathbb{Z}$ and $p = 2$.

(2) If $r = 1$ then $\ell^1(G)$ has the SP and the RNP.

QUESTION: *It is not clear to us if $A_p^r(G)$ has the DPP if $r > \max\{p, p'\}$ in case G is discrete, or even if $G = \mathbb{Z}$ and $p = 2$.*

If G is abelian non compact then $A_2(G)$ does not have the SP, since $\hat{G}$ is non discrete. This is substantially improved in the main result, namely

THEOREM A: *Let G be a noncompact locally compact group.*

(1) $\forall 1 < r \leq \infty$, $A_p^r(G)$ does not have the SP
Yet $A_p^1(G)$ has the SP if G is discrete.

IDEA OF THE PROOF

For $G = \mathbb{Z}$, the Fourier algebra is simply the Fourier transform of $L^1(\mathbb{T})$. For finite r , the space $A_2^r(\mathbb{Z})$ is the graph space of the Fourier coefficient map

$$f \mapsto \hat{f}$$

inside $L^1(\mathbb{T}) \oplus \ell^r(\mathbb{Z})$, with norm

$$\|f\|_1 + \|\hat{f}\|_r.$$

The monomials z^n are weakly null in $L^1(\mathbb{T})$, while the unit vectors e_n are weakly null in ℓ^r for $1 < r < \infty$. Thus the corresponding elements of $A_2^r(\mathbb{Z})$ are weakly null. On the dual side, the coordinate functionals are weakly null because they come from the weakly null coordinate basis of $\ell^{r'}$. Their pairings with the matching monomials stay equal to one, which is exactly the standard obstruction to the Dunford–Pettis property.

PROOF

Theorem 1. *Let $1 < r < \infty$. Then $A_2^r(\mathbb{Z})$ does not have the Dunford–Pettis property.*

Proof. Use normalized Haar measure on $\mathbb{T}$. Under the Fourier transform, $A_2(\mathbb{Z})$ is isometric to $L^1(\mathbb{T})$, and

$$A_2^r(\mathbb{Z}) = \{\widehat{f} : f \in L^1(\mathbb{T}), \widehat{f} \in \ell^r(\mathbb{Z})\}$$

with norm $\|\widehat{f}\|_{A_2^r} = \|f\|_1 + \|\widehat{f}\|_{\ell^r}$. Equivalently, it is the graph subspace

$$X_r = \{(f, \widehat{f}) : f \in L^1(\mathbb{T}), \widehat{f} \in \ell^r(\mathbb{Z})\} \subset L^1(\mathbb{T}) \oplus_1 \ell^r(\mathbb{Z}).$$

For $n \in \mathbb{Z}$, let $x_n \in X_r$ correspond to the Fourier monomial z^n ; in graph notation,

$$x_n = (z^n, e_n),$$

where e_n is the n th coordinate vector of $\ell^r(\mathbb{Z})$. The sequence (z^n) is weakly null in $L^1(\mathbb{T})$: for every $g \in L^\infty(\mathbb{T}) \subset L^1(\mathbb{T})$, the Riemann–Lebesgue lemma gives

$$\int_{\mathbb{T}} z^n g(z) dm(z) \longrightarrow 0$$

as $|n| \rightarrow \infty$. Also (e_n) is weakly null in $\ell^r(\mathbb{Z})$ since $1 < r < \infty$. Hence (x_n) is weakly null in the graph subspace X_r .

Let $\varphi_n \in X_r^*$ be the restriction to X_r of the n th coordinate functional on the ℓ^r summand:

$$\varphi_n(f, \widehat{f}) = \widehat{f}(n).$$

The coordinate basis of $\ell^{r'}(\mathbb{Z}) = (\ell^r(\mathbb{Z}))^*$ is weakly null because $1 < r' < \infty$. Therefore (φ_n) is weakly null in X_r^* , being the image of a weakly null sequence under the adjoint of the inclusion $X_r \hookrightarrow L^1(\mathbb{T}) \oplus_1 \ell^r(\mathbb{Z})$.

But the pairings do not tend to zero:

$$\varphi_n(x_n) = e_n(n) = 1 \quad (n \in \mathbb{Z}).$$

Thus there are weakly null sequences $(x_n) \subset X_r$ and $(\varphi_n) \subset X_r^*$ whose scalar pairings stay equal to one. This is precisely a failure of the Dunford–Pettis property. $\square$

Theorem 2. *For the high-exponent question with $G = \mathbb{Z}$ and $p = 2$:*

$$A_2^r(\mathbb{Z}) \text{ has the DPP} \iff r = 1 \text{ or } r = \infty$$

within the source's range $1 \leq r \leq \infty$.

Proof. For $1 < r < \infty$, this is the previous theorem. For $r = 1$, the source already records that $A_p^1(G) = \ell^1(G)$ for discrete G , hence has the Schur property and therefore the DPP. For $r = \infty$, the definition gives $A_2^\infty(\mathbb{Z}) = A_2(\mathbb{Z})$. Since $A_2(\mathbb{Z}) \cong L^1(\mathbb{T})$ and $L^1(\mu)$ has the Dunford–Pettis property for every measure μ , the endpoint $r = \infty$ has the DPP. $\square$

LIMITATIONS

This settles the $G = \mathbb{Z}$, $p = 2$ Dunford–Pettis subquestion. It does not settle whether $A_2^r(\mathbb{Z})$ has the Radon–Nikodym property for finite $r > 2$, nor the corresponding questions for $G = \mathbb{R}$, nor the source conjecture that $A_2^r(G)$ has the DPP for connected semisimple Lie groups with finite center and $r > 2$.

NOVELTY AND VERIFICATION NOTES

- Local run indexes were searched for 2006.13021, $A_2^r(\mathbb{Z})$, Dunford–Pettis, $A_p^r(G)$, and related Fourier-algebra phrases. No prior packet for this source or exact question was found.
- Bounded web search on 2026-06-29 for “ $A_2^r(\mathbb{Z})$ Dunford Pettis Property”, “ $A_p^r(G)$ Dunford–Pettis Z ”, and the source title found only the source paper, not a later answer.
- The proof uses only the standard Fourier identification $A_2(\mathbb{Z}) \cong L^1(\mathbb{T})$, the Riemann–Lebesgue lemma, reflexivity of ℓ^r for $1 < r < \infty$, and the classical fact that $L^1(\mu)$ has the Dunford–Pettis property.

HUMAN REVIEW RECOMMENDATION

Review as a short, likely valid scoped full solution. The key point to check is that weak convergence in the graph subspace follows from weak convergence in the ambient direct sum; this is immediate by Hahn–Banach extension of functionals from the subspace.

REFERENCES

- [1] E. E. Granirer, *Geometric Properties of some Banach Algebras related to the Fourier algebra on Locally Compact Groups*, arXiv:2006.13021, 2020. <https://arxiv.org/abs/2006.13021>
- [2] J. Diestel, *A survey of results related to the Dunford–Pettis property*, Contemporary Mathematics 2, American Mathematical Society, 1980.